Lesson 5: Dictionaries: Look It Up

Module 2 | 60 minutes | dictionaries, keys, values | Grades 7–11 · ages 12–16

Objective: Store information as key and value pairs in a dictionary, look values up, add and update them, loop over them, and build a class quiz bank.

Success criteria

- I can make a dictionary of key: value pairs.
- I can look up, add and update a value by its key.
- I can check a key exists with in before using it.
- I can loop over a dictionary with .items().

Key vocabulary

- **dictionary**: A collection of key: value pairs inside { }.
- **key**: The label you look up, like "chai". Keys are unique.
- **value**: The information stored with a key, like 10.
- **.items()**: Gives every key and value pair, for looping.
- **KeyError**: The error when you ask for a key that is not in the dictionary.
- **.get(key, default)**: Reads a value safely, giving the default if the key is missing.

You will need

- Projector with Kitabash (lesson video, Python Lab)
- Devices: 1 per learner, with a keyboard
- Printed worksheet (optional)
- Project files: canteen_menu.py

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game: Human Phone Book**	Play Human Phone Book (unplugged). The class shouts “KeyError!” for a missing key. One learner updates a value.
5 min	**Hook and video**	Show the hook. Chat poll: “What will print?” Play the video with captions; pause at “Count with a dictionary” and ask what count will hold.
10 min	**Live demo and build-along**	Run the live demo including the class fruit vote, then everyone types the canteen menu. Thumbs check: “Did pizza get the Sorry message?”
8 min	**Challenge, predict and match**	Canteen price checker, the class election prediction and the dictionary match. Quick competition: fastest correct match wins.

15 min	**Quiz bank in breakout pairs**	Each pair agrees on a topic, writes 5 questions on paper, then both build the quiz bank. Partners play each other's quiz in the chat.
6 min	**Debug race**	Race to fix "the broken report card". The winner explains both bugs.
7 min	**Quiz show-and-tell**	Two learners host their quiz bank live and the class answers in the chat. Celebrate the highest class score.
4 min	**Wrap-up and exit ticket**	Fist-to-five on the success criteria. Exit ticket: change the notebook price.

Unplugged warm-up

Human Phone Book (5 min game): five learners each hold a "key" card (a made-up nickname like Captain, Speedy, Spinner) and secretly write a "value" (a favourite snack). The class calls out keys and the holder reveals the value. Calling a key nobody holds gets "KeyError!" shouted by the class. Then one learner changes their value: that is updating a key.

Teaching the idea

A dictionary stores key: value pairs inside curly brackets. We read a value with square brackets and its key, like menu["chai"]. Assigning to a key adds it if it is new or replaces the value if it exists. in checks whether a key exists, which avoids KeyError. Looping with .items() gives each key and value. Today the quiz from Level 1 becomes data: questions live in a dictionary and one loop asks them all.

Analogy: A dictionary is like a phone book or the contacts on a phone: you search by the name (key) to find the number (value). You never search by position, and two contacts cannot have exactly the same name.

Live demo: 1) Type capitals = {"Kerala": "Thiruvananthapuram", "Punjab": "Chandigarh"} and print(capitals["Punjab"]). 2) Try capitals["punjab"] and read the KeyError aloud. 3) Add capitals["Goa"] = "Panaji" and print the dictionary. 4) Make the canteen menu, loop with for item, price in menu.items(): and print each line. 5) Show order in menu with input. 6) Do the fruit vote count: ask learners to shout fruits, type them into a list, and count them with the dictionary pattern.

Questions to ask

Ask	Listen for
What is the difference between a key and a value?	You look up by the key; the value is what is stored with it.
What happens with menu["idli"] = 30 if idli is already there?	The old price is replaced with 30.
How do you avoid a KeyError?	Check with if key in dict: or use .get(key, default).
What does for item, price in menu.items(): give you each time?	One key in item and its value in price.

Why is a dictionary good for a quiz?	Each question is linked to its answer and one loop can ask them all.

Differentiation

Support
Give the quiz bank starter with all 5 questions filled in; learners write only ask() and the loop. A printed card shows the shape { "key": value, "key": value }.

Stretch
Add a scores dictionary for two players taking turns, or shuffle the questions with random.shuffle(list(questions.items())).

Common misconceptions

Misconception	How to fix it
Dictionaries are looked up by position, like lists	You look up by key, not by number position.
"Maths" and "maths" are the same key	Keys must match exactly, including capitals.
for x in my_dict gives the values	It gives the keys. Use .items() for pairs or .values() for values.
A dictionary can have two pairs with the same key	Keys are unique; assigning again replaces the value.

Assessment

Use the Python Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
prices = {"pen": 10, "notebook": 40}. Write one line that changes the notebook price to 45.

Homework
Make a “family favourites” dictionary at home: ask three people for their favourite food (use first names or nicknames only) and write a program that prints each pair and counts how many people chose sweets.

Answer key — Kitabash activities

1. Python Lab: canteen price checker (perform · python lab)

Goals checked: Chai costs Rs 10 · Samosa costs Rs 15 · Pizza is not on the menu · Uses a dictionary and if

One solution:

```
menu = {"samosa": 15, "vada pav": 20, "chai": 10}
order = input("What would you like? ").strip().lower()
if order in menu:
    print(f"That will be Rs {menu[order]}, please.")
else:
    print("Sorry, that is not on the menu.")
```

2. Predict the output: class election (perform · code output)

3. Match each piece of dictionary code to what it does (perform · matching)

menu["chai"] → Reads the value for the key chai · menu["chai"] = 12 → Adds chai or changes its value · "chai" in menu → Checks if the key chai exists · menu.items() → Gives every key and value pair · menu.get("tea", 0) → Reads a value, or 0 if the key is missing

4. Project: class quiz bank (perform · python lab)

Goals checked: Asks 5 questions · Wrong answers score 0 · Shows the final score · Uses a dictionary, a loop and a function

Example program:

```
# Class Quiz Bank
# Made by: Aarav

questions = {
    "Which planet is closest to the Sun?": "mercury",
    "How many sides does a hexagon have?": "6",
    "What is the capital of Rajasthan?": "jaipur",
    "Which gas do plants take in from the air?": "carbon dioxide",
    "What is 9 x 7?": "63",
}

def ask(question, answer):
    reply = input(question + " ").strip().lower()
    if reply == answer:
        print("Correct!")
        return 1
    print(f"Not quite. The answer is {answer}.")
    return 0

print("=== CLASS QUIZ BANK ===")
print(f"There are {len(questions)} questions. Let's go!")
score = 0
for question, answer in questions.items():
    score += ask(question, answer)

print(f"Final score: {score} out of {len(questions)}")
percent = score * 100 // len(questions)
print(f"That is {percent}%")
if percent == 100:
    print("Quiz master!")
elif percent >= 60:
    print("Great job!")
else:
    print("Keep revising and try again!")
```

5. Debug it: the broken report card (probe · python lab)

Goals checked: Prints the Maths mark · Prints every subject · Prints the total

One solution:

```
marks = {"Maths": 85, "Science": 92, "English": 68}
marks["Hindi"] = 74
print("Maths mark:", marks["Maths"])
total = 0
for subject, mark in marks.items():
    print(f"{subject}: {mark}")
    total += mark
print("Total:", total)
```

Answer key — worksheet

1. **Key words:** dictionary = A collection of key: value pairs inside { }. · key = The label you look up, like "chai". Keys are unique. · value = The information stored with a key, like 10. · .items() = Gives every key and value pair, for looping.
2. **Plan your program:** Goals: Chai costs Rs 10; Samosa costs Rs 15; Pizza is not on the menu; Uses a dictionary and if. One solution: menu = {"samosa": 15, "vada pav": 20, "chai": 10} order = input("What would you like? ").strip().lower() if order in menu: print(f"That will be Rs {menu[order]}, please.") else: print("Sorry, that is not on the menu.")
3. **Predict the output: class election:** A. 3 5
4. **Match each piece of dictionary code to what it does:** menu["chai"] → Reads the value for the key chai · menu["chai"] = 12 → Adds chai or changes its value · "chai" in menu → Checks if the key chai exists · menu.items() → Gives every key and value pair · menu.get("tea", 0) → Reads a value, or 0 if the key is missing
5. **Plan your program:** Goals: Prints the Maths mark; Prints every subject; Prints the total. One solution: marks = {"Maths": 85, "Science": 92, "English": 68} marks["Hindi"] = 74 print("Maths mark:", marks["Maths"]) total = 0 for subject, mark in marks.items(): print(f"{subject}: {mark}") total += mark print("Total:", total)

Kitabash original curriculum.